

MANGALORE, Australia

WMO: 948740

Lat: 36.8886S

Lon: 145.1858E

Elev: 141

StdP: 99.64

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-0.2	0.9	-2.2	3.2	14.1	-0.9	3.6	10.5	10.2	11.8	9.3	11.2	1.2	210	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.6	36.2	18.6	33.8	18.5	31.7	18.0	22.0	28.0	20.6	28.1	19.6	27.8	5.5	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.6	15.5	23.6	18.6	13.7	22.4	17.1	12.4	21.7	65.0	27.6	60.0	28.0	56.4	27.7	33.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	9.0	8.2	DB	-2.3	40.2	0.9	2.0	-2.9	41.7	-3.4	42.9	-3.9	44.0	-4.5	45.5	(4)
(5)			WB	-2.3	25.6	1.0	3.6	-3.0	28.2	-3.5	30.3	-4.0	32.4	-4.7	35.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.7	22.2	21.7	19.0	14.9	11.3	8.8	8.1	8.8	10.8	13.8	17.3	19.6
	DBStd	5.87	4.00	3.63	3.57	3.06	2.54	2.20	2.01	2.19	2.66	3.33	3.82	3.94
	HDD10.0	203	0	0	0	2	14	48	64	51	19	4	1	0
	HDD18.3	1743	9	10	34	108	217	286	317	296	225	147	63	31
	CDD10.0	1906	378	328	280	150	56	12	6	13	44	122	220	298
	CDD18.3	405	129	104	55	7	0	0	0	0	1	6	33	70
	CDH23.3	5127	1619	1188	608	79	1	0	0	0	6	142	502	982
	CDH26.7	2317	837	552	227	11	0	0	0	0	1	42	196	452
Wind	WSAvg	4.0	5.0	4.7	4.2	3.6	3.3	3.3	3.4	3.5	3.8	4.2	4.6	4.8
Precipitation	PrecAvg	574	38	34	31	40	58	58	60	66	59	50	46	41
	PrecMax	972	116	216	110	165	135	169	119	149	178	175	150	123
	PrecMin	259	0	1	2	4	4	12	8	11	12	3	1	1
	PrecStd	142	28	43	23	31	36	34	32	34	37	36	35	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.6	38.2	35.2	28.1	21.7	17.2	15.6	18.6	23.7	30.9	34.6	38.0
		MCWB	19.6	20.2	18.4	15.9	13.6	11.7	11.2	13.2	14.3	16.5	17.3	18.5
	2%	DB	36.7	34.7	31.6	25.5	19.3	15.2	13.9	16.3	20.7	27.2	31.5	34.3
		MCWB	19.0	19.2	17.8	15.6	13.3	11.5	10.5	11.6	14.0	15.0	16.8	17.9
	5%	DB	33.9	32.3	29.2	23.6	17.6	14.1	12.9	14.8	18.6	24.2	28.7	31.5
		MCWB	18.8	18.8	17.4	15.0	12.7	10.9	9.9	10.7	13.1	14.5	16.4	17.2
	10%	DB	31.4	30.2	26.8	21.6	16.2	13.0	12.0	13.5	16.9	21.5	26.0	28.9
		MCWB	18.2	18.2	16.7	14.3	12.2	10.3	9.4	9.9	12.2	13.7	15.9	16.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.3	24.7	22.0	19.2	16.0	13.7	12.7	13.9	16.8	18.6	21.0	23.4
		MCDB	28.1	28.1	24.8	23.7	18.0	14.8	13.8	17.3	19.6	25.7	28.0	26.6
	2%	WB	21.8	21.7	20.0	17.1	14.6	12.3	11.3	12.3	15.0	16.7	19.1	20.1
		MCDB	29.3	27.8	26.8	21.4	17.1	14.1	12.9	15.4	18.7	22.6	26.3	27.3
	5%	WB	20.4	20.4	18.7	16.0	13.5	11.4	10.5	11.2	13.7	15.6	17.8	18.8
		MCDB	29.4	28.0	25.7	21.0	16.4	13.4	12.2	14.1	17.9	21.8	25.0	27.1
	10%	WB	19.2	19.3	17.7	14.9	12.7	10.7	9.7	10.3	12.4	14.3	16.7	17.7
		MCDB	29.0	27.5	24.7	20.8	15.7	12.6	11.5	13.0	16.2	20.7	23.6	26.8
Mean Daily Temperature Range	5% DB	MDBR	14.6	13.8	12.8	11.9	9.6	8.3	8.0	9.4	11.2	13.0	13.5	14.5
		MCDBR	19.3	17.8	16.7	15.1	11.5	9.6	9.1	11.3	14.2	18.3	18.5	19.3
		MCWBR	5.7	5.4	5.7	6.9	6.4	6.1	6.0	7.0	8.2	8.1	6.4	6.4
	5% WB	MCDBR	15.2	13.9	13.1	12.3	9.4	7.7	7.6	10.2	12.3	14.5	13.9	14.9
		MCWBR	6.3	6.3	6.2	6.8	5.8	5.5	5.4	6.7	7.8	7.8	6.4	6.9
Clear-Sky Solar Irradiance	taub	0.344	0.344	0.321	0.314	0.301	0.293	0.291	0.301	0.326	0.329	0.342	0.335	
	taud	2.494	2.503	2.564	2.562	2.567	2.581	2.585	2.541	2.460	2.476	2.474	2.511	
	Ebn at Noon	990	966	951	897	852	831	856	899	929	971	985	1003	
	Edh at Noon	114	108	94	83	71	66	69	83	102	110	115	113	
All-Sky Solar Radiation	RadAvg	7.53	6.62	5.37	3.79	2.54	1.94	2.12	3.01	4.28	5.70	6.68	7.45	
	RadStd	0.42	0.36	0.26	0.20	0.17	0.21	0.12	0.23	0.35	0.47	0.36	0.37	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)		+0.51	N/A	N/A	+0.96	N/A	N/A	-18	-129	+185	+66

Nomenclature: See separate page